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Statement of Work For Bioforensics R&D technical support
[REDACTED]**1.0 INTRODUCTION**

The Biological Technology Center (BTC) under the Central Intelligence Agency's Directorate of Science and Technology (DS&T) Department of Advanced Technologies and Programs (ATP) is supporting forensic microbiology research and development (R&D) efforts for the intelligence community (IC) in coordination with and with support from the Intelligence Technology Innovation Center (ITIC). Currently capabilities are robust for detection and identification of microbial pathogens of BW concern from a variety of matrices. However, detection and identification are only superficial levels of forensic analysis. Forensic comparative analyses carry with them a more stringent set of scientific requirements. To meet the forensic requirements significant R&D investments must be pursued. BTC is supporting work in this regard and leveraging partnerships across government agencies wherever possible to support and further develop robust forensic microbiology capability.

The goal of this project is to leverage the sequencing, functional genomics and bioinformatics capabilities that the National Institutes of Allergy and Infectious Disease (NIAID/NIH) has funded at the Institute for Genomic Research (TIGR). NIAID has a pre-existing relationship with TIGR to provide comprehensive genomic, bioinformatics, and functional genomics reagents, resources and data to the scientific community for conducting research on organisms considered agents of bioterrorism. TIGR has also sought to foster a specific capability in forensic microbiology by building a partnership with [REDACTED]. The collaborative efforts between TIGR and [REDACTED] are producing a research Program in Integrated Bio-threat Forensics-Systems Biology. Many of the underlying basic science questions relevant to forensic microbiology can be examined at the unclassified level through comparative genomic analyses and studies on agents of bio-terrorism supported by NIAID at TIGR to generate reagents, technologies and data for the scientific community. To date no such collaborative efforts exist anywhere else at the [REDACTED] level with this specific goal. This project will use BTC funds to leverage the pre-existing efforts funded by NIAID at TIGR to support BTC research requirements and add capacity to perform functional comparative genomics studies in a way that is of benefit to both NIAID and BTC. Unclassified basic scientific research on questions of forensic relevance such as technique validation strategies, mutational stability of genetic loci, and data interpretation will be used to augment and complement other BTC directed R&D efforts.

2.0 SCOPE

BTC requests that NIAID contract with TIGR under the NIAID Pathogen Functional Genomics Resource Center (PFRGC) contract ([REDACTED]) to conduct work to support BTC relevant forensic research needs and focus comparative genomic analysis of organisms considered agents of bioterrorism and use emerging genomic

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technologies to study these organisms at a whole genome level not focused on one gene at a time. This collaboration between BTC and NIAID not only meets BTC scientific requirements but is also directly related to the mission of NIAID and fits well within the goals and capacities of the NIAID-funded PFGRC efforts. In carrying out this work TIGR should leverage any appropriate preexisting academic collaborations. The work shall encompass validation studies for existing bio-threat analytic methods as directed by BTC to include test and evaluation of comparative genomic analysis methods such as re-sequencing micro-arrays for bacterial pathogens using samples provided by NIAID and obtained from other government organizations when applicable with BTC approval and facilitation as necessary. Finally, using model pathogen systems that are scientifically comparable to existing classes of current bacterial threat agents (with BTC approval), conduct work to further development of comparative analysis analytic tools as forensic analytic tools, validate interpretive criteria to include in addition to traditional sequence-based genotyping approaches.

3.0 TASKS

Task 3.1: Within 1 month of including budget and transfer IAA agreement, the contractor shall produce a detailed work execution plan for comparative analyses and validation studies using DNA resequencing microarrays specific for *Francisella tularensis* with the possibility to expand the studies to include other bioterror threat agents. The plan shall be presented to the government Sponsor (BTC) and NIAID during a Contract Initiation Review (CIR) meeting. The validation studies must include work to examine the mutational stability of genetic loci of forensic utility present in the genomes of these pathogens. NIAID will provide sample sets for this purpose that leverage other government collaborations and BTC will facilitate NIAID obtaining sample sets from other US Government entities as appropriate. This task should comprise approximately 15% of the overall effort under this contract. BTC shall take the lead to ensure NIAID success with respect to obtaining the required sample sets.

Task 3.2: Within two months of IAA award the contractor shall start the work outlined in Task 3.1 focusing initial studies first on *Francisella tularensis* and proceeding to the other pathogens. BTC will aid NIAID in obtaining the appropriate samples sets for conducting the analyses. By the end of the contract period the contractor shall produce a report that describes all of the results and interpretations from these studies. The report should also include extensive discussions on the impact of variations in mutational rates seen at different genetic loci and how this might impact forensic genetic comparison to include the limitations of interpretations. The government sponsors shall be kept informed of the progress on this task through quarterly reports and bimonthly meetings during which progress, data, and any project relevant difficulties are discussed and resolved with the government COTR. With the mutual consent of the government sponsors the frequency of the bimonthly meetings can be adjusted as progress on

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the effort warrants. This task represents 70% of the overall effort expected under this contract.

Task 3.3: In parallel to tasks 3.1 and 3.2, using model organisms that are appropriate surrogates for the organisms in task 3.1, develop analytic and interpretive criteria for the exclusion, inclusion, and 'matching' of bacterial samples from known and questioned sources based on forensic genetic comparisons. Where appropriate leverage information developed from Tasks 3.1 and 3.2. Describe the genetic loci informative and important to the criteria to include: what is known about these loci, the locus importance for forensic comparison to include the effects that loci genetic stability may have on forensic comparative analysis. If insufficient relevant research data is available for the organisms selected (and the actual pathogens listed in task 3.1) develop a research plan that would generate the relevant data and test the hypothetical approaches developed. Leverage relevant academic collaborations as necessary. The selection rationale for surrogate organisms and the draft research plan are to be presented to the government Sponsors in the form of a report that incorporates responses to Sponsor feedback, concerns and questions. This task should represent 15% of the overall effort under this contract.

4.0 DELIVERABLES

4.1 For Task 3.1: Within 1 month of IAA NIAID shall provide a detailed and fully explained work execution plan as outlined under task 3.1. The contractor shall provide 5 hardcopies of this plan for government review and one electronic copy in CD-rom format. All electronic documents shall be provided in Microsoft Word format and copies of any presentations provided in Microsoft powerpoint.

4.2 For task 3.2: The contractor shall provide the government sponsors with a quarterly report discussing the results and implications of all analyses produced as part of this task. The contractor shall also provide all raw data to the government as requested. The contractor shall provide 5 hardcopies of these reports for government review and one electronic copy in CD-rom format. All electronic documents shall be provided in Microsoft Word format and copies of any presentations prepared during bimonthly meetings shall be provided in Microsoft powerpoint.

4.3 For Task 3.3: The contractor shall provide a detailed and fully explained research plan as outlined under task 3.3. The contractor shall provide 5 hardcopies of this plan for government review and one electronic copy in CD-rom format. All electronic documents shall be provided in Microsoft Word format and copies of any presentations provided in Microsoft powerpoint. The deliverable for this task is a written plan that could be executed at a future date if the BTC is interested in pursuing it further.

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4.4 Progress, Status, and Management Reports.

The contractor shall meet quarterly with the government sponsor to discuss progress on the tasks and provide the government with Quarterly progress reports. These reports and the other deliverables are to be marked within the header and footer as "UNCLASSIFIED//FOUO", whereby FOUO carries the meaning 'For Official (government) Use Only.' Upon completion of the the studies the contractor shall provide a final report which contains a detailed discussion of all results and analyses associated with all the tasks described in this SOW. All deliverables are to be sent by FEDEX to:



Alternatively the progress reports and deliverables can be provided directly in electronic format to the government sponsors at the quarterly meeting.

5.0 Presentation Material - Meeting Minutes/Conference Notes.

Quarterly progress reviews shall be held and attended by the representatives of the sponsors CIA and NIAID and appropriate government sponsor approved PFRGC-TIGR staff.

6.0 Markings/Distribution of Reports

6.1 Data Rights. The CIA considers all technical data developed using funds transferred to NIAID under this Interagency Agreement to be sensitive, but unclassified, with respect to National Security for the development of forensic attribution capabilities in bio-defense, and as such should not be used, modified, performed, displayed, released or disclosed, in whole or in part in any manner and for any purpose whatsoever without the CIA's permission. This information should be treated as For Official Use Only (FOUO). Accordingly, NIAID shall include the following clause in any contract for the development of technical data using CIA funds provided under this IAA. If a contractor refuses to accept this clause, or substantially similar terms, NIAID will promptly bring such refusal to the CIA's attention and will not proceed with the award of the contract without authorization of the CIA. NIAID will not authorize the contractor to use or disclose such data without the prior permission of the CIA. Unless required by law, regulation or court order, NIAID will not disclose such information in its possession or control without first seeking the CIA's permission. NIAID will make reasonable efforts to promptly inform the CIA of any request or requirement for disclosure.

6.2 Distribution/Release Limitation Statements. All data shall be protected from disclosure in accordance with the markings contained on each document and as described in the following text. All other information relating to the items to be delivered or

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services to be performed under this delivery order may not be disclosed by any means without prior approval by the CIA.

DISTRIBUTION STATEMENT B: Distribution authorized to U.S. Government Agencies only upon Central Intelligence Agency approval.

DESTRUCTION NOTICE - For unclassified, limited documents, destroy by any method that will prevent disclosure of contents or reconstruction of the document.

7.0 PERIOD OF PERFORMANCE

The Period of Performance for the base effort is one base year beginning at the date of execution of the IAA between CIA and NIAID and up to two one year option periods subsequent to the agreement execution. The base year will begin upon award of the contract with exercise of option year periods at the discretion of the Sponsor.

8.0 SECURITY

8.1 The transfer of funds from the Central Intelligence Agency and the contractual relationship of the CIA with NIAID will be UNCLASSIFIED//FOUO.

8.2 No data or other deliverables from this project shall be distributed, published, briefed or otherwise used without the express approval of the sponsor, CIA (DST/ATP/BTC).

9.0 NIAID Points of Contact for this Effort.

Technical Project Manager:

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National Institute of Allergy and Infectious Diseases
6610 Rockledge Drive, [REDACTED]
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